Algorithm1: C-IFS-FWZIC Method

Step 1: Define the sustainable Sensing Parameters Attributes:
identify $SSPA[i]$

Step 2: Structured Expert Judgment:
Define EX[i]

Define Ques, NS
m ← length (EX)
For i in {1..m}
if EX(i) is true then EX(i) ← Ques(i)
endif
endfor

Step 3: Building the Expert Decision Matrix (EDM):
Initialize $\mathrm{EDM[i,j]} \leftarrow \mathrm{E} \uplus SSPA$
J ← length ($SSPA$)
m ← length (EX)
For j in {1..J}
For i in {1..m}
$$\mathrm{EDM[i,j]} \leftarrow \mathrm{NS}(\mathrm{EX_{ij}}/SSPA_{\mathrm{ij}})$$
endfor
endfor

Step 4: Application of C-IFS Membership Function:
For j in {1..J}
For i in {1..m}
$$\mathrm{C-IFS\ EDM[i,j]} \leftarrow \mathrm{EDM[i,j]}$$
endfor
endfor

Step 5: Apply C-IFS Aggregation Equation:
For j in {1…J}

$SSPA_{\mathrm{j}} \leftarrow \sum_{\mathrm{i=1}}^{\mathrm{m}} \mathrm{NS}\ (\breve{\mathrm{E}}_{\mathrm{ij}}/SSPA_{ij})$

$$\mu(C_i), v(C_i)\rangle = \left(\frac{\sum_{j=1}^{K_i} m_{i,j}}{k_i}, \frac{\sum_{j=1}^{K_i} n_{i,j}}{k_i}\right)$$

endfor

Step 6: Calculate the Maximum Radius Lengths
For j in {1…J}

$SSPA_{\mathrm{j}} \leftarrow \sum_{\mathrm{i=1}}^{\mathrm{m}} \mathrm{NS}\ (\breve{\mathrm{E}}_{\mathrm{ij}}/SSPA_{ij})$

$$r_i = max_{1 \le j \le k_i} \sqrt{(\mu(C_i) - m_{i,j})^2 + (v(C_i) - n_{i,j})^2}$$

endfor

Step 6.1: Create the C-IFS
For j in {1…J}

$SSPA_{\mathrm{j}} \leftarrow \sum_{\mathrm{i=1}}^{\mathrm{m}} \mathrm{NS}\ (\breve{\mathrm{E}}_{\mathrm{ij}}/SSPA_{ij})$
$A_r = \{(C_i, \mu(C_i), v(C_i); r) | C_i \in W\} = \{(C_i, O_r(\mu(C_i), v(C_i))) | C_i \in W\}$
endfor

Step 7: Calculate the Optimistic Decision Matrix and Pessimistic Decision Matrix

For j in {1…J}

$SSPA_{\mathrm{j}} \leftarrow \sum_{\mathrm{i=1}}^{\mathrm{m}} \mathrm{NS}\ (\breve{\mathrm{E}}_{\mathrm{ij}}/SSPA_{ij})$

$$Q^{od} = \begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ A_n \end{array} \begin{array}{c} \begin{array}{ccc} \mathrm{A1} & \dots\ \mathrm{A2} & \dots\ A_m \end{array} \\ \begin{bmatrix} ((\mu_{11} + r_{11}, v_{11} - r_{11})) & ((\mu_{12} + r_{12}, v_{12} - r_{12})) \cdots & ((\mu_{1m} + r_{2m}, v_{1m} - r_{2m})) \\ ((\mu_{21} + r_{21}, v_{21} - r_{n1})) & ((\mu_{22} + r_{22}, v_{22} - r_{22})) \ddots & ((\mu_{2m} + r_{2m}, v_{2m} - r_{2m})) \\ ((\mu_{n1} + r_{n1}, v_{n1} - r_{n1})) & ((\mu_{n2} + r_{22}, v_{n2} - r_{n2})) \cdots & ((\mu_{nm} + r_{nm}, v_{nm} - r_{nm})) \end{bmatrix} \end{array}$$

$$Q^{o_d} == \begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ A_n \end{array} \begin{array}{c} \begin{array}{ccc} \text{A1} & \dots \text{A2} & \dots A_m \end{array} \\ \begin{bmatrix} ((\mu_{11} - r_{11}, v_{11} + r_{11})) & ((\mu_{12} - r_{12}, v_{12} + r_{12})) \cdots & ((\mu_{1m} - r_{2m}, v_{1m} + r_{2m})) \\ ((\mu_{21} - r_{21}, v_{21} + r_{n1})) & ((\mu_{22} - r_{22}, v_{22} + r_{22})) \ddots & ((\mu_{2m} - r_{2m}, v_{2m} + r_{2m})) \\ ((\mu_{n1} - r_{n1}, v_{n1} + r_{n1})) & ((\mu_{n2} - r_{22}, v_{n2} + r_{n2})) \cdots & ((\mu_{nm} - r_{nm}, v_{nm} + r_{nm})) \end{bmatrix} \end{array}$$

endfor

Step 8: Calculate the Weights W_k for each DM

Step 9: Applying the IFWA Operator Equation

For j in {1…J}

$SSPA_j \leftarrow \sum_{i=1}^{m} NS\ (\check{E}_{ij}/SSPA_{ij})$

$$IFWA\ (A_1, A_2, \dots, A_n) = \left(\sum_{j=1}^{n} w_j A_j\,, \left(1 - \prod_{i=1}^{n} (1 - \mu_j)^{w_i}, \prod_{j=1}^{n} v_j^{w_j} \right) \right),$$

endfor

Step 10: Determine the maximum radius lengths $R^w = (r_j^w)_{1xn}$ based on the attribute weight matrices of all DMs

Step 11: Calculate the Optimistic Attribute Weight Matrix and the Pessimistic Attribute Weight Matrix

For j in {1…J}

$SSPA_j \leftarrow \sum_{i=1}^{m} NS\ (\check{E}_{ij}/SSPA_{ij})$

$$Q^{o_w} = \begin{array}{c} A_1 \\ A_2 \\ \vdots \\ A_n \end{array} \begin{bmatrix} ((\mu_1 + r_1, v_1 - r_1)) \\ ((\mu_2 + r_2, v_2 - r_2)) \\ ((\mu_n + r_n, v_n - r_n)) \end{bmatrix}$$

$$Q^{p_w} = \begin{array}{c} A_1 \\ A_2 \\ \vdots \\ A_n \end{array} \begin{bmatrix} ((\mu_1 - r_1, v_1 + r_1)) \\ ((\mu_2 - r_2, v_2 + r_2)) \\ ((\mu_n - r_n, v_n + r_n)) \end{bmatrix}$$

endfor

Step 12: obtain the weighted optimistic decision matrix and the weighted pessimistic decision matrix

For j in {1…J}

$SSPA_j \leftarrow \sum_{i=1}^{m} NS\ (\check{E}_{ij}/SSPA_{ij})$

$$\psi_{ij}^{o} = q_j^{o_w} \otimes q_{ij}^{o_d}$$

$$\psi_{ij}^{p} = q_j^{p_w} \otimes q_{ij}^{p_d}$$

endfor

Step 13: Find the Score Values
For j in {1..J}

$$S(A) = \mu_A - v_A$$

$$H(A) = \mu_A + v_A$$

endfor

Step 14: Find the Final Weight

For j in {1..J}

$$w_{cj}^{i} = \frac{w_{cj}^{i} \times w_{fj}^{i}}{\sum_{j=1}^{n} w_{cj}^{i} \times w_{fj}^{i}}$$

endfor
Endfor
Endfor

Algorithm 2: ARAS
Input: Alternatives, Attributes, and Attributes Weights
Output: Select the Best Alternative

Step1: Formulated and Normalized Decision Matrix and Calculate the Weighted Normalized Decision Matrix

Step 2: Determine the Priority of Alternatives to define Normalized Weighted Matrix
Identify rA[j]

Identify IoTRTMD A[i]

n← length (A)
m ← length (IoTRTMD A)
For j in {1..J}
For i in {1..m}
if IoTRTMD A (i) is true then

$$X_{ij} = \frac{X_{ij}}{\sum_{i=0}^{m} X_{ij}}$$

$$X_{ij} = \frac{1}{X_{ij}^{*}}$$

Endif
Endfor
Endfor

Step 3: Calculate the Maximum Sum of Weight

$$\sum_{j=1}^{n} w_j = 1$$

Step 4: Calculate the Normalized Weighted Values of each Attribute

J ← length (rA)
m ← length (CBMI A)
For j in {1..J}
For i in {1..m}

$$X_{ij} = X_{ij} W_j$$

Endfor
Endfor

Step 5: Calculate the Values of the Optimality Function

For j in {1..J}
For i in {1..m}

$$S_i = \sum_{j=1}^{n} X_{ij}$$

Endfor
Endfor

Step 6:	Determine the Utility Degree of Alternative
	
	Identify rA[j]
	Identify IoTRTMD A[i]
	n← length (A)
	For j in {1..J}
	For i in {1..m}
	
	if IoTRTMD A (i) is true then
	$$K_i = \frac{S_i}{S_0}, i = 0, m$$
	Endfor
	Endfor

TABLE A.I
DECISION MATRIX RESULTS

Attributes / Alternatives	Temperature	Relative Humidity	Gas Composition	Location	Light Intensity	Pressure	Weight	Microbial Concentration	Vibration	Air Velocity
IoTRTMD 1	1	1	1	0	0	0	0	0	1	0
IoTRTMD 2	1	1	1	0	1	0	0	0	0	0
IoTRTMD 3	1	1	1	0	0	0	0	0	0	0
IoTRTMD 4	1	1	1	0	0	1	0	0	0	0
IoTRTMD 5	1	1	0	0	0	0	1	0	0	0
IoTRTMD 6	0	0	0	0	0	0	0	1	0	0
IoTRTMD 7	1	0	0	1	0	0	0	0	0	0
IoTRTMD 8	0	0	1	0	0	0	0	0	0	0
IoTRTMD 9	1	1	0	0	0	0	1	0	0	0
IoTRTMD 10	1	1	0	0	0	0	0	0	0	0
IoTRTMD 11	0	0	1	0	0	0	0	0	0	0
IoTRTMD 12	1	1	1	0	0	0	0	0	0	0
IoTRTMD 13	1	1	1	0	0	0	0	0	0	0
IoTRTMD 14	1	0	0	0	0	0	0	0	0	0
IoTRTMD 15	1	1	0	0	0	0	0	0	0	0
IoTRTMD 16	1	1	1	0	0	0	0	0	0	0
IoTRTMD 17	1	0	0	0	0	0	0	0	0	0
IoTRTMD 18	1	0	0	0	0	0	0	0	0	0
IoTRTMD 19	1	1	0	0	0	0	0	0	0	0
IoTRTMD 20	1	1	0	0	0	0	0	0	0	0
IoTRTMD 21	1	1	0	0	0	0	0	0	0	0
IoTRTMD 22	1	0	0	0	0	0	0	0	0	0
IoTRTMD 23	1	1	1	0	0	0	0	0	0	0
IoTRTMD 24	1	0	0	0	0	0	0	0	0	0
IoTRTMD 25	1	1	0	0	0	0	0	0	0	0
IoTRTMD 26	1	1	1	0	0	1	0	0	0	0
IoTRTMD 27	1	1	0	0	1	0	0	0	0	0
IoTRTMD 28	1	1	0	0	0	0	0	0	0	0
IoTRTMD 29	0	0	0	1	0	0	1	0	0	0
IoTRTMD 30	1	1	1	0	0	0	0	0	0	0
IoTRTMD 31	1	1	1	0	0	0	0	0	0	0
IoTRTMD 32	1	1	0	0	1	0	0	0	0	0
IoTRTMD 33	1	0	0	0	0	0	0	0	0	0
IoTRTMD 34	1	0	0	0	0	0	0	0	0	0
IoTRTMD 35	1	1	0	0	0	0	0	0	0	0
IoTRTMD 36	1	1	1	0	0	0	0	0	0	0
IoTRTMD 37	1	1	1	0	0	0	0	0	0	0
IoTRTMD 38	1	1	0	1	1	0	0	0	0	0
IoTRTMD 39	1	1	0	1	0	0	0	0	0	0
IoTRTMD 40	1	1	1	0	1	0	0	0	0	0
IoTRTMD 41	0	0	0	0	0	0	0	1	0	0
IoTRTMD 42	1	0	0	0	0	0	0	0	0	0
IoTRTMD 43	1	0	0	0	0	0	0	0	0	0
IoTRTMD 44	1	1	0	0	1	0	0	0	0	0
IoTRTMD 45	1	1	0	0	1	0	0	0	0	0
IoTRTMD 46	1	0	0	0	0	0	0	0	0	0
IoTRTMD 47	1	1	1	0	0	0	0	0	0	0
IoTRTMD 48	1	1	1	0	0	0	0	0	0	0
IoTRTMD 49	1	1	0	0	0	1	0	0	0	1
IoTRTMD 50	1	0	0	0	0	0	0	0	0	0
IoTRTMD 51	1	0	0	0	0	1	0	0	0	1
IoTRTMD 52	0	0	0	0	0	0	1	0	0	0
IoTRTMD 53	1	1	0	1	0	0	0	0	1	0
IoTRTMD 54	1	1	0	0	0	0	0	0	0	0